

The Association of Consumption of Fruits/Vegetables with Decreased Risk of Glaucoma among Older African American Women in the Study of Osteoporotic Fractures

Purpose To explore the association between consumption of fruits and vegetables and the presence of glaucoma in older African American women. **Design** Cross-sectional study. **Methods** Disc photographs and suprathreshold visual fields were obtained from the 662 African American participants in the Study of Osteoporotic Fractures. Masked, trained readers graded all discs, and two glaucoma specialists reviewed photos and visual fields. The Block Food Frequency Questionnaire assessed food consumption. Relationships between selected fruit/vegetable/nutrient consumption and glaucoma were evaluated using logistic regression models after adjusting for potential confounders. **Results** After excluding women missing Food Frequency Questionnaire and disc data, 584 African American women (88.2% of total African American cohort) were included. Glaucoma was diagnosed in at least one eye in 77 subjects (13%). Women who ate 3 or more servings/day of fruits/fruit juices were 79% (odds ratio [OR]=0.21; 95% confidence interval [CI]: 0.08–0.60) less likely to have glaucoma than women who ate less than one serving/day. Women who consumed more than 2 servings/week of fresh oranges (OR=0.18; 95%CI: 0.06–0.51) and peaches (OR=0.30; 95%CI: 0.13–0.67) had a decreased odds of glaucoma compared to those consuming less than one serving/week. For vegetables, >1 serving/week compared to ≤1 serving/month of collard-greens/kale decreased the odds of glaucoma by 57% (OR=0.43; 95%CI: 0.21–0.85). There was a protective trend against glaucoma in those consuming more fruit/fruit juices (p=0.023), fresh oranges (p=0.002), fresh peaches (p=0.002), and collard greens/kale (p=0.014). Higher consumption of carrots (p=0.061) and spinach (p=0.094) also showed some associations. Individual nutrient intake from food sources found protective trends with higher intakes of vitamin A (p=0.011), vitamin C (p=0.018), and α -carotene (p=0.021), and close to statistically significant trends with β -carotene (p=0.052), folate (p=0.056), and lutein/zeaxanthin (p=0.077). **Conclusion** Higher intake of certain fruits and vegetables high in Vitamins A and C and carotenoids may be associated with a decreased likelihood of glaucoma in older African American women. Randomized controlled trials are needed to determine whether the intake of specific nutrients changes the risk of glaucoma.